

Computer Vision: Digital Images & Color Models

Interactive Learning Activities • Observe → Experiment → Explain

How to use this page

This interactive exercise is based on the lesson topics: **image representation, RGB, colors and light, color spaces, and white balancing.**

Important: These activities are about discovery. Make your observations first, then explain what you think happened.

Key idea: A digital image is represented using pixels, and an RGB pixel stores three numerical values. But the numbers measured by a camera are affected by light and viewing conditions.

Task 1 — Meet the Pixel

Image Representation + RGB

Choose a colorful image and inspect five pixels using an image editor or pixel-inspection tool.

Pixel	R	G	B	Predicted Color
1	255	217	84	Yellow
2	7	119	87	Green

3	227	1	28	Red
4	68	145	189	Blue
5	251	52	159	Pink

Which pixel has the highest overall intensity? Why?

Pixel 1 (Yellow), because it had higher RGB values

What happens when the three RGB values are approximately equal?

The color will have no distinct hue or saturation

Task 2 — Can You Beat the Camera?

Light + Pixel Intensity

Photograph a white, gray, and black object together. If possible, create a second photo where the lighting differs across the objects.

Prediction

Brightest object

White Paper

Darkest object

Black Case

Observation

What did the pixel values tell you?

Pixel values provide information of each pixel. Higher RGB values generally mean a brighter pixel, while lower values mean a darker pixel.

Object	R	G	B	Approx. intensity
White	234	235	237	235.3

Gray	130	137	121	129.3
Black	21	37	27	28.3

Can a black object have a higher pixel intensity than a white object? Explain.

Yes, it can. Pixel intensity depends on the lighting and image conditions, not just the object's actual color. For example, a black object under strong light may produce higher pixel values than a white object in a very dark or shadowed area.

Does a pixel value always tell us the actual color of an object?

No. A pixel value represents the color recorded by the camera or sensor at that location. It can be affected by lighting, shadows, reflections, camera settings, and surrounding colors.

Task 3 — Build a Color

RGB Color Model

Use the sliders or number fields to create colors by changing only R, G, and B.



RGB: (255, 217, 83)

Target color you tried

Yellow

What happened when you increased only one channel?

for yellow, I had to max out Red and Green to get Yellow. But when I max Blue also, the color starts turning white

Predict before experimenting

What do you expect when all three RGB values are increased equally?

It becomes white and the brightness goes up

What do you expect when all three RGB values are decreased equally?

It starts to turn black and darken

Task 4 — RGB vs. HSV Detective

Color Spaces

Your lecture introduces RGB and HSV as different ways of mathematically representing color.

Color

RGB

HSV

Yellow

255,217,84

60,90%,78.43%

Repeat for several colors in your notes or on a separate table.

Which HSV component seems related to the basic color?

Hue is related to the basic color. In my color, the Hue is 60, which represents yellow.

Which component seems related to color strength/purity?

Saturation represents the color's strength or purity. In my color, saturation is 90%, so the color is quite vivid.

Which component seems related to brightness/value?

Value represents brightness. In my color, value is 78.43%, meaning the color is relatively bright.

You need to detect red objects. Would RGB or HSV be easier to work with? Explain your reasoning.

HSV would generally be easier to work with. The Hue component directly represents the basic color, so red objects can be detected by looking for pixels within a certain red hue range. Saturation and Value can also be used to filter out gray/dull pixels and very dark

Task 5 — The White Balance Mystery

White Balance

Photograph the same white object under different lighting conditions.

Photo	Lighting	How does white appear?	R	G	B
A	Under the tabl	It looks like a l	155	142	126
B	Under regular	I can accurate	235	234	230
C	Under a warm	It starts to hav	220	198	174

The object is physically the same. Why can its RGB values change?

The RGB values is drastically affected by the surroundings of the object when the photo was taken. Especially the light quality and type of light that is used to light up the object

Why might a camera need white balance?

The camera needs a white balance so that the camera can adjust itself to accurately that the correct representation of the white object regardless of the surroundings affecting the photo

How does this relate to last week's shadow activity?

In last week's shadow activity we had a similar situation where the same object had different RGB colors due to lighting. The white balance of a camera aims to solve this by adjusting the colors of the photo such that white objects actually look white.

Task 6 — Think Like a Computer Vision System

Synthesis

Choose a photograph containing several objects or colors. Imagine that you are the computer.

Choose one pixel. What RGB information does it contain?

GB: 190,56,53

Based on its RGB values, what color does it appear to represent?

Red

Choose another pixel from the same object. Are its values exactly the same? If not, why?

RGB: 51, 188, 229 (Blue). No its not the same, it has a different RGB values

Would a human necessarily interpret the two pixels as different-colored objects?

Yes as humans can interpret the visual difference between the colors

What problem could changing illumination create for object recognition?

How the object is illuminated drastically affects how a computer can see the object, not allowing it to see the true color

Think like a CV system: A human can immediately attach meaning to a scene. A computer vision system starts from numerical image data.

Final Challenge — The Color Investigator

Choose one photograph you took yourself and investigate it as a Computer Vision student.

A. Image Representation

B. RGB: record three pixels

C. Light: where does lighting
affect the image?

Overview

1. Meet the Pixel

2. Beat the Camera

3. Build a Color

4. RGB vs HSV

5. White Balance

photo

D. Color Space: how could
another color space represent
it?

1: HSV(33.2°, 18.9%,
97.6%)
2: HSV(22.0°, 56.7%,
82.4%)
3: HSV(216.4°, 23.7%,
92.5%)

E. White Balance: is there a
color cast? What might cause
it?

Yes, the warm light
affects how the white
earpod looks

F. Human vs. Computer: what
does a human understand
that one pixel cannot tell us?

we somehow know it is
actually white because
of the context we have.

Final Reflection

Complete the statement:

"A digital image is not simply a picture. It is..."

A digital image is not simply a picture. It is a collection of pixels, where each pixel contains numerical values that computers can understand

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